AN1149

Design A Load Sharing System Power Path Management with Microchip's Stand-Alone Li-Ion Battery Charger

Author: *Brian Chu*
Microchip Technology Inc.

INTRODUCTION

Batteries often serve as the main energy source for portable electronic devices. Although they depend on batteries, portable consumer electronic products, such as GPS and multi-media players, often consume energy directly from ac-dc wall adapter or accessory power adapter (or "Auto Adapter") when the battery is low or the device is in a stationary mode.

Relatively high energy density and maintenance free make Lithium-Ion (Li-Ion) batteries popular in the portable consumer electronic products. Li-Ion batteries are the example of the portable power source in this applications note.

Depending on the product design or local government regulations, rechargeable batteries are often charged inside the handheld devices or from battery charger cradles. Due to the safety concerns or design concepts in some products, the batteries are required to be removed from the portable device and charge directly from the battery charger cradle.

However, most of time, batteries are designed to be recharged inside the devices while the devices are still in the operational mode. An end-user can extend run time while refilling the energy back to the battery for the next mobile action. The adapter now has to power up the device while charging the battery. The battery can deliver energy to the system load when the power source is absent as depicted in Figure 1.

Portable electronic devices play an important role in a person's daily life from grocery shopping to personal data assistant and have dramatically changed the way people live and work. With the emerging technologies that are available today, portable electronic designers are trying to integrate more features into smaller and lighter form-factors while extending the system run times.

This application note is intended to assist product designers by taking advantage of using Microchip's fully integrated simple Li-Ion battery charge management controllers with common directional control to build a system and battery load sharing circuirty. The solutions are ideal for use in cost-sensitive applications that can also accelerate the product time-to-market rate.

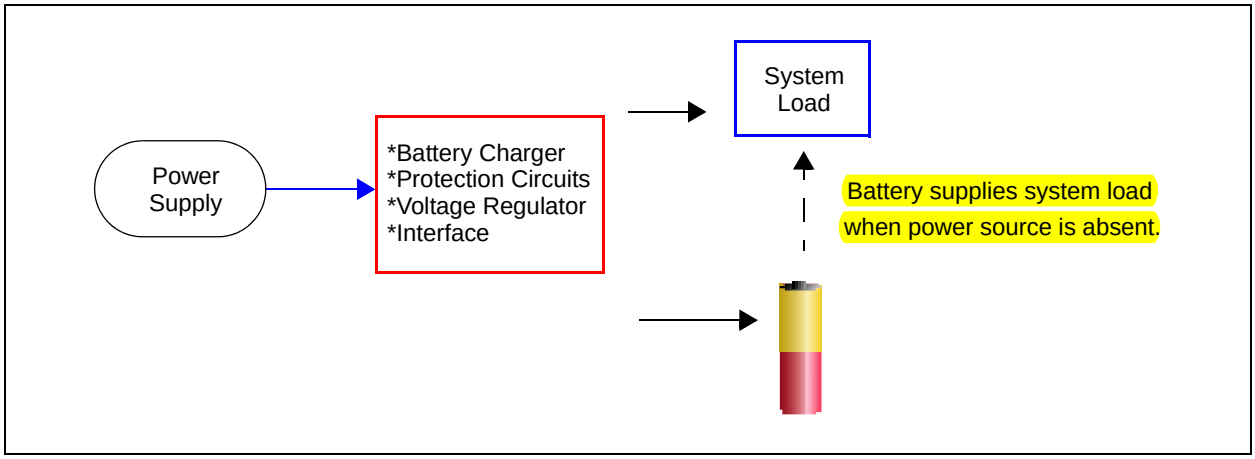


FIGURE 1: *Typical Portable Power Source.*

© 2007 Microchip Technology Inc.

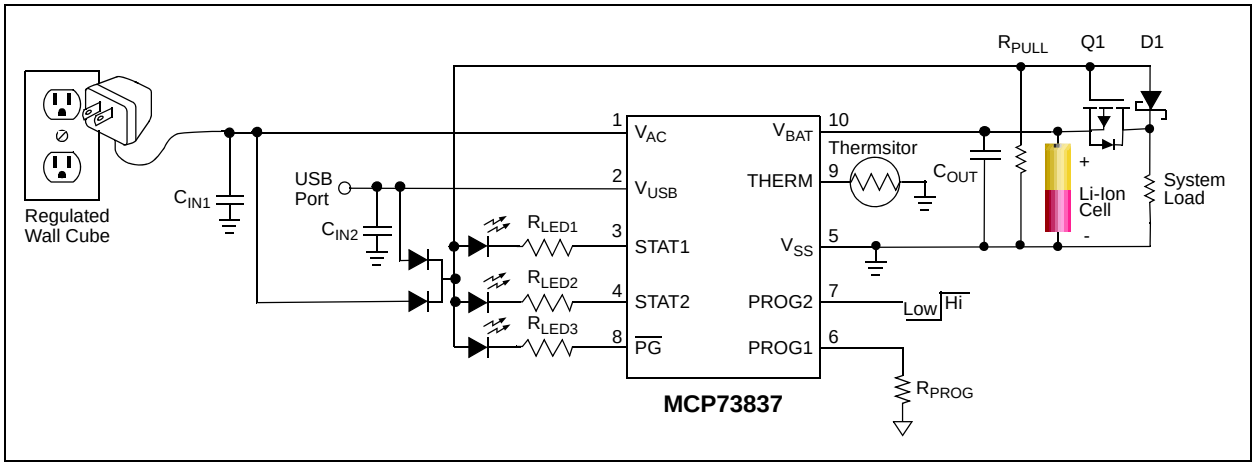


FIGURE 2: *Typical System and Battery Load Sharing Application.*

DESCRIPTION

This applications note shows how to design a simple system load sharing with Microchip's popular *"Advanced Stand-Alone Li-Ion / Li-Polymer Battery Charge Management Controller with Autonomous AC-Adapter or USB-Port Source Selection"* for cost-sensitive applications.

References to documents that treat these subjects in more depth and breadth have been included in the **"Reference"** section.

BATTERY CHARGER AND SYSTEM LOAD DESIGN SPECIFICATIONS AND APPLICATION DESCRIPTION

The example system load consumes a maximum 500 mA when all applications are running at the same time. A 950 mAh rated Li-Ion battery is required to operate the example portable system for approximately two hours during intensive load operations.

The input power should supply the system load and charge battery when a battery is present in the system. When input power source is removed, the system is supported by the battery. When the system load and the battery draw more energy that the supply current can offer, the system load has priority over the battery charger.

Design Specifications

- Input Voltage Range:
  - 4.5V - 6.5V from ac-dc adapter (1A)
  - 5V from USB port (100 mA/500 mA)
  - 3V - 4.2V from 1-cell Li-Ion battery (950 mAh)
- Constant Charge Current:
  - 0.5C (The battery manufacturer recommended value)
  - 100 mA / 500 mA (Charge from USB port)
- Precondition Current:
  - 0.1C or recommend value
- Termination Current:
  - 0.07C
- Charge Status and Power Good Indicators
- Safety Timer: Turn charger off after 6 Hours before termination

© 2007 Microchip Technology Inc.

LI-ION (LITHIUM-ION) / LI-POLYMER (LITHIUM POLYMER) BATTERIES

There are some important attributes when selecting a battery for an application:

1. Internal Resistance,
2. Operational Load Current,
3. Energy Density (Size & Weight),
4. Charge/Discharge Cycles (Life Cycle),
5. Capacity (dominates the operational duration without external power source present).

Like the most engineering work, these key attributes do not exist in the same technology. There is always a trade-off between them when selecting the battery chemistry for a portable application. Refer to Microchip's AN1088 *"Selecting the Right Battery System for Cost-Sensitive Portable Applications While Maintaining Excellent Quality"* for the details of battery chemistry comparisons.

Li-Ion batteries have played an important role in today's portable world because of the advantages in high energy density, low maintenance requirement, relatively low self discharge rate and higher cell voltages. 1-cell Li-Ion batteries especially enjoy the largest share of the Li-Ion battery market while 1-cell and 2-cell applications are available in more than 70% of total available market. An example of the single-cell Li-Ion battery charger and system load sharing power path circuit is demonstrated in this application note.

Li-Polymer batteries which are also recognized as Li-Ion Polymer batteries are similar in terms of chemistry with Li-Ion batteries. Li-Polymer can be charged as same algorithm as Li-Ion batteries because of the similar characteristics. The flexible form-factors and very low profile to fit inside the compact applications make it an ideal candidates for MP3 Players and Mobile Phones.

Note:	The major drawbacks of Li-Ion batteries are higher initial cost and aging effect. Li-Ion batteries age over time regardless the number of cycles that have been reached. A Protection circuit is required for Li-Ion batteries to prevent over voltage during charge cycle and under voltage during discharge cycle; overcurrent as well in both directions.

Batteries usually occupy a considerable amount of space and weight in today's portable devices. The energy density for each chemistry dominates the size and weight for the battery pack. Li-Ion has advantages in both energy density weight and energy density volume among other available battery technologies.

SELECT BATTERY CHARGE MANAGEMENT CONTROL CIRCUIT

The emerging semiconductor technologies shorten the design cycles and simplify design methods for the consumer product designers by integrating circuits into a single chip. The first step is to decide to design a custom charge control management circuit or adapt a standalone charge IC.

Stand-Alone Charge Management Controller

Microchip's MCP73837 *"Advanced Stand-Alone Li-Ion / Li-Polymer Battery Charge Management Controller with Autonomous AC-Adapter or USB-Port Source Selection"* is selected to complete the design because it dramatically reduces the software/hardware design time and simplify the PCB layout. The selected MCP73837 is regulated at 4.20V and programmed at 500 mA for ac-dc wall adapter. The general features of MCP73837/8 are listed below:

- High Accuracy Preset Voltage Regulation: ± 0.5%
- Available Voltage Regulation Options:
  - 4.20V, 4.35V, 4.4V or 4.5V
- Complete Linear Charge Management Controller
  - Autonomous Power Source Selection
  - Integrated Pass Transistors
  - Integrated Current Sense
  - Integrated Reverse Discharge Protection
- Constant Current / Constant Voltage Operation with Thermal Regulation
- Selectable USB-Port Charge Current:
  - 100 mA maximum (Logic Low) / 500 mA maximum (Logic High)
- Programmable AC-Adapter Charge Current:
  - 15 mA - 1000 mA
- Two Charge Status Outputs
- Power-Good Monitor:
  - MCP73837
- Timer Enable:
  - MCP73838
- Automatic Recharge
- Automatic End-of-Charge Control
  - Selectable Charge Termination Current Ratio
  - Selectable Safety Timer Period
- Preconditioning of Deeply Depleted Cells
- Battery Cell Temperature Monitor
- UVLO (Under Voltage Lockout)
- Automatic Power-Down when Input Power Removed
- LDO (Low-Dropout) Linear Regulator Mode
- Minimum External Components Required

© 2007 Microchip Technology Inc.

- Numerous Selectable Options Available for a Variety of applications:
  - Refer to Datasheet Section 1.0 “Electrical Characteristics for Selectable Options”
  - Refer to the Datasheet “Product Identification System” for Standard Options
- Temperature Range: -40°C to +85°C
- Packaging:
  - 10-Lead 3 mm x 3 mm DFN
  - 10-Lead MSOP

Note:	Refer to MCP73837/8 datasheet (DS22071) for detail descriptions.

Common Cathode Diode

A common cathode diode is applied here to drive LEDs and supply system load when either source from ac-dc adapter or USB-port. It prevents reverse current feeding into the other source.

A common cathode diode can be left out of the design if an automatic switch between ac-dc adapter and USB port feature is not required or a different charge IC which does not have autonomous dual power source selection is adapted to the design. Figure 3 depicted a closer look of the common cathode diode that is used in the design.

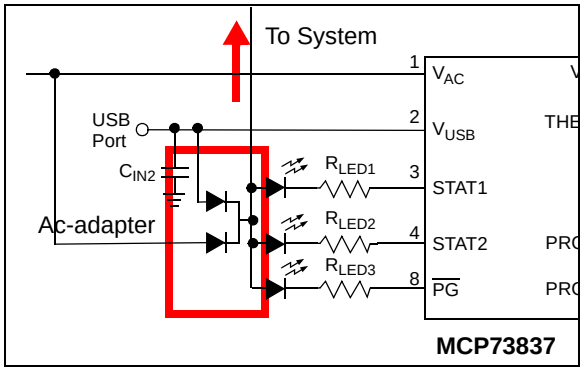


FIGURE 3: *Common Cathode.*

SELECT THE SYSTEM LOAD SUPPLY CIRCUIT

Some designers may think an additional circuit is not required and simply connect a system load to the battery cell. The current flows to system when the sources are removed and randomly dump current into battery pack when the extra current is available.

Here are few reasons that the system load is not recommended to be directly connected to the battery terminals:

1. The charge may never end - Most Li-Ion battery chargers are based on CC-CV (Constant Current and Constant Voltage) mode. The termination is based on the ratio of output current and constant current. If the system constant draw current from the charger, the charge current never will meet the termination value.

Note:	MCP73811/2 Li-Ion battery charge management controllers with no auto-termination may be a good solution for the type of applications that are designed to simply tie system load to Li-Ion battery.

2. The total system current is limited by charge current because the charger will deliver total system and battery charging current through the output pin. This problem may be feasible for some applications that run on constant current, but not recommended.

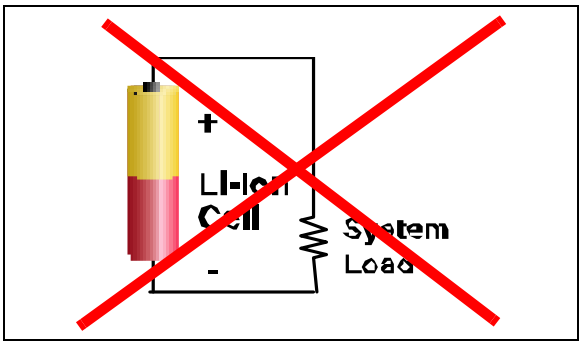


FIGURE 4: *Do not just connect the load to battery.*

Select The Pull Down Resistor

The R_{PULL} in Figure 5 represents pull down resistor to make sure that the P-Ch MOSFET (Q1) turns on when input sources are removed. When the input sources are absent, the R_{PULL} pulls the gate to zero allowing current to flow out of the battery.

R_{PULL} value can be any reasonable resistor. However, R_{PULL} value should not be too small. The small R_{PULL} wastes unnecessary current when input sources are present. 100 kΩ R_{PULL} is recommended in this design which consumes about 50 µA when V_{IN} = 5V.

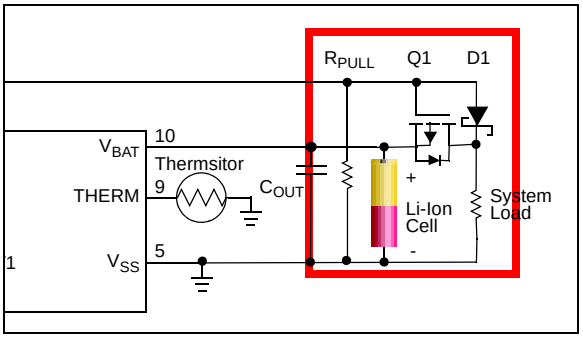


FIGURE 5: *Current Directional Control Circuit.*

© 2007 Microchip Technology Inc.

Select The MOSFET

The nature of the MOSFET makes it the best candidate for current direction control. A P-Channel MOSFET is selected to complete this circuit. Figure 6 depicts when V_{IN} is available, the gate of Q1 is High. Q1, which blocks the current from Li-Ion battery is turned off and allows current to support the system load from V_{IN} when charging the battery from V_{BAT} pin. When the gate of Q1 is Low, Q1 turns on and allows the Li-Ion battery to supply the system as shown in Figure 7. The MCP73837 VBAT pin also is disabled when V_{IN} is absent.

Note:	It is important to select proper a proper gate threshold voltage range, so the MOSFET will be turned on.

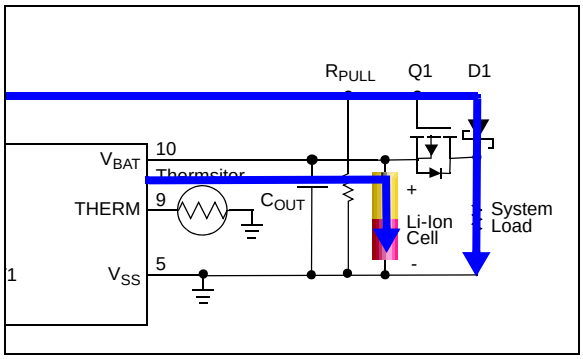


FIGURE 6: *Q1 is Off when gate is High and no current flows from the battery cell to system load.*

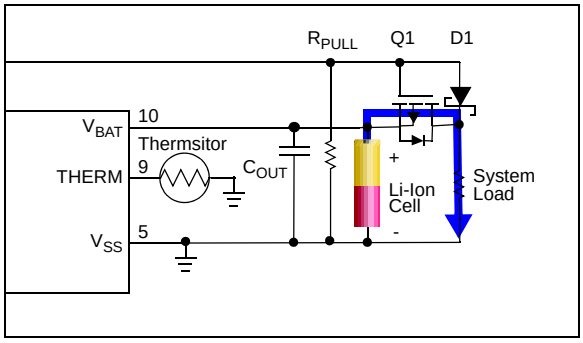


FIGURE 7: *Q1 is ON when gate is Low and current flows from the battery cell to system load.*

Select The Diode

A diode is required to prevent reverse current to the power source as depicted in Figure 6 and Figure 7. Selecting the right diode can minimize the leakage current and forward voltage drop from power source to the system load. A schottky diode, which has lower forward voltage drop is recommended.

Note:	Average Forward Current has to be rated greater than the maximum system load current for the application.

Co-packaged MOSFET + Schottky Diode

Semiconductor manufacturers provide MOSFET and Schottky Diode in one small package to save board space and cost. A typical SO-8 packaged low forward voltage drop schottky diode and power P-Ch MOSFET was used for demonstration in this application note.

CHARGE PROFILE WITH SYSTEM LOAD

A complete system load and battery power path management circuit, which depicted in Figure 2 was designed for demonstration purpose in this application note. The system load was setup at a constant 500 mA rate. A deeply depleted 950 mAh Li-Ion battery was used and charged by Microchip's MCP73837 *"Advanced Stand-Alone Li-Ion / Li-Polymer Battery Charge Management Controller with Autonomous AC-Adapter or USB-Port Source Selection"*. A fast charge current of USB_{High} was selected to charge 450 mA in Constant Current Mode. The MCP73837/8 was designed to charge at a typical 450 mA constant current when USB_{High} is selected and guaranteed not exceed 500 mA limit when high-power USB port is available.

Note:	The USB (Universal Serial Bus) Specification clearly defined that a device may be either low-power at 100 mA loads or high-power, consuming up to 500 mA loads. All devices default to low-power. The transition to high-power is under software control. It is the responsibility of software to ensure adequate power is available before allowing devices to consume high-power. The number of unit loads a device can draw is an absolute maximum, not an average over time. (Designers should obtain the latest design guide line and detail information from the USB-IF if USB peripherals is going to be implemented in a project.)

© 2007 Microchip Technology Inc.

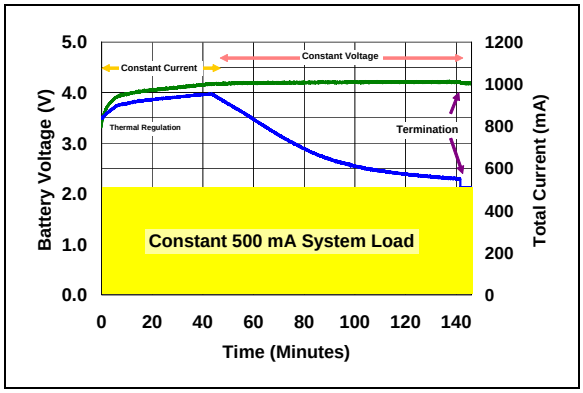


FIGURE 8: *450 mA Constant Charge Current Li-Ion Battery Charge Profile With Constant 500 mA System Load.*

Stage 1: Preconditioning - Preconditioning is employed to restore charge to deeply depleted cells. When the cell voltage is below designed threshold voltage, the cell is charged with a constant current of 0.1C maximum. This period is hard to see from Figure 8 because the VBAT rises above 3V in very short period of time and enters the Constant Current (Fast Charge) mode.

Stage 2: Constant Current - Once the cell voltage has risen above the preconditioning threshold, the charge current is increased to perform fast charging. The fast charge current should not be more than 1C. A fast charge current of 450 mA (~ 0.5C) is used in this example. The thermal foldback period demonstrates temperature regulation by limiting current during Fast Charge Period which also improve the reliability and prolong the life of charger IC.

Stage 3: Constant Voltage - Fast charge ends, and the Constant Voltage mode is initiated when the cell voltage reaches 4.2V (Or the desired available option, which is available to order via Microchip for MCP73837/8). In order to maximize capacity, the voltage regulation tolerance should be better than ±1%. MCP73837/8 provides a ±0.5% superior voltage regulation tolerance to deliver maximum battery runtime after each completed charge cycle.

Stage 4: Termination

Charging is typically terminated by one of two methods: minimum charge current or a timer (or a combination of the two). The MCP73837/8 employs the combination termination method. Figure 8 shows that the minimum current is reached before time-out occurs. MCP73837/8 monitors the charge current during the constant voltage stage and terminates the charge when the charge current diminishes below approximately 0.07C (Other termination ratio options are available for this device from Microchip Technology).

Note:	It is not recommended to continue to trickle charge Lithium-Ion batteries.

Charging in this manner replenishes a deeply depleted battery in roughly 140 minutes at 0.5C mA. Advanced chargers employ additional safety features. For example, charge is suspended if the cell temperature is outside a specified window, typically 0°C to 45°C.

After 140 minutes, Figure 8 demonstrates that the power supply still supports a solid 500 mA system load when charge termination occurs and the battery charger went into standby mode. During this standby mode, the MCP73837 continuos to monitor the V_{BAT} and will recharge the Li-Ion battery once it drops 150 mV below regulated V_{BAT} voltage.

DISCHARGE THE LI-ION BATTERY

The input power source was removed when a full charge cycle was completed. The P-Ch MOSFET was turned on to supply the system load with 0.53C and discharged the 950 mA Li-Ion battery as depicted in Figure 9. The termination duration is load dependent and the Figure 9 also shows the Li-Ion battery was not able to deliver 500 mA after 105 minutes. With approximately 0.5C discharge rate, the time should last about 2 hours. The main reason that the remain 15 minutes are not available from this experiment is because the remaining capacity level is not enough to support 500 mA.

Note:	Fully depleted a Li-Ion battery may degrade its life cycles and should be avoided.

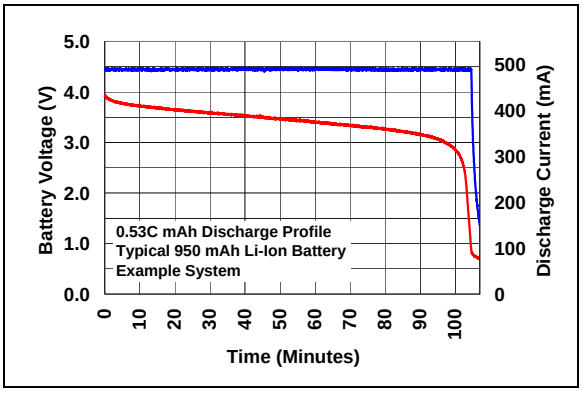


FIGURE 9: *500 mA Discharge Profile when V_{IN} is removed.*

© 2007 Microchip Technology Inc.

CONCLUSION

System and battery load sharing power path management circuits are very common in portable applications. Adapting this simple design wisely can dramatically reduce the total system cost and product developing time in order to take advantages of using a fully integrated battery charge management controller.

REFERENCES

[1] *"Lithium Batteries"*, Gholam-Abbas Nazri and Gianfranco Pistoia Eds.; Kluwer Academic Publishers, 2004.

[2] *"Portable Electronics Product Design and Development"*, Bert Haskell; McGraw Hill, 2004.

[3] "AN1088, *"Selecting the Right Battery System for Cost-Sensitive Portable Applications. While Maintaining Excellent Quality"*, Brian Chu;
Microchip Technology Inc., DS01088, 2007.

[4] MCP73837/8 Data Sheet, *"Advanced Stand-Alone Li-Ion / Li-Polymer Battery Charge Management Controller with Autonomous AC-Adapter or USB-Port Source Selection"*,
Microchip Technology Inc., DS22071, 2007.

© 2007 Microchip Technology Inc.

NOTES:

© 2007 Microchip Technology Inc.

Note the following details of the code protection feature on Microchip devices:

- Microchip products meet the specification contained in their particular Microchip Data Sheet.
- Microchip believes that its family of products is one of the most secure families of its kind on the market today, when used in the intended manner and under normal conditions.
- There are dishonest and possibly illegal methods used to breach the code protection feature. All of these methods, to our knowledge, require using the Microchip products in a manner outside the operating specifications contained in Microchip's Data Sheets. Most likely, the person doing so is engaged in theft of intellectual property.
- Microchip is willing to work with the customer who is concerned about the integrity of their code.
- Neither Microchip nor any other semiconductor manufacturer can guarantee the security of their code. Code protection does not mean that we are guaranteeing the product as "unbreakable."

Code protection is constantly evolving. We at Microchip are committed to continuously improving the code protection features of our products. Attempts to break Microchip's code protection feature may be a violation of the Digital Millennium Copyright Act. If such acts allow unauthorized access to your software or other copyrighted work, you may have a right to sue for relief under that Act.

Information contained in this publication regarding device applications and the like is provided only for your convenience and may be superseded by updates. It is your responsibility to ensure that your application meets with your specifications. MICROCHIP MAKES NO REPRESENTATIONS OR WARRANTIES OF ANY KIND WHETHER EXPRESS OR IMPLIED, WRITTEN OR ORAL, STATUTORY OR OTHERWISE, RELATED TO THE INFORMATION, INCLUDING BUT NOT LIMITED TO ITS CONDITION, QUALITY, PERFORMANCE, MERCHANTABILITY OR FITNESS FOR PURPOSE**.** Microchip disclaims all liability arising from this information and its use. Use of Microchip devices in life support and/or safety applications is entirely at the buyer's risk, and the buyer agrees to defend, indemnify and hold harmless Microchip from any and all damages, claims, suits, or expenses resulting from such use. No licenses are conveyed, implicitly or otherwise, under any Microchip intellectual property rights.

Trademarks

The Microchip name and logo, the Microchip logo, Accuron, dsPIC, KEELOQ, KEELOQ logo, microID, MPLAB, PIC, PICmicro, PICSTART, PRO MATE, rfPIC and SmartShunt are registered trademarks of Microchip Technology Incorporated in the U.S.A. and other countries.

AmpLab, FilterLab, Linear Active Thermistor, Migratable Memory, MXDEV, MXLAB, SEEVAL, SmartSensor and The Embedded Control Solutions Company are registered trademarks of Microchip Technology Incorporated in the U.S.A.

Analog-for-the-Digital Age, Application Maestro, CodeGuard, dsPICDEM, dsPICDEM.net, dsPICworks, dsSPEAK, ECAN, ECONOMONITOR, FanSense, FlexROM, fuzzyLAB, In-Circuit Serial Programming, ICSP, ICEPIC, Mindi, MiWi, MPASM, MPLAB Certified logo, MPLIB, MPLINK, PICkit, PICDEM, PICDEM.net, PICLAB, PICtail, PowerCal, PowerInfo, PowerMate, PowerTool, REAL ICE, rfLAB, Select Mode, Smart Serial, SmartTel, Total Endurance, UNI/O, WiperLock and ZENA are trademarks of Microchip Technology Incorporated in the U.S.A. and other countries.

SQTP is a service mark of Microchip Technology Incorporated in the U.S.A.

All other trademarks mentioned herein are property of their respective companies.

© 2007, Microchip Technology Incorporated, Printed in the U.S.A., All Rights Reserved.

Printed on recycled paper.

QUALITY MANAGEMENT SYSTEM
CERTIFIED BY DNV
ISO/TS 16949:2002

Microchip received ISO/TS-16949:2002 certification for its worldwide headquarters, design and wafer fabrication facilities in Chandler and Tempe, Arizona; Gresham, Oregon and design centers in California and India. The Company's quality system processes and procedures are for its PIC® MCUs and dsPIC® DSCs, KEELOQ® code hopping devices, Serial EEPROMs, microperipherals, nonvolatile memory and analog products. In addition, Microchip's quality system for the design and manufacture of development systems is ISO 9001:2000 certified.

WORLDWIDE SALES AND SERVICE

AMERICAS

Corporate Office
2355 West Chandler Blvd.
Chandler, AZ 85224-6199
Tel: 480-792-7200
Fax: 480-792-7277
Technical Support:
http://support.microchip.com
Web Address:
www.microchip.com

Atlanta
Duluth, GA
Tel: 678-957-9614
Fax: 678-957-1455

Boston
Westborough, MA
Tel: 774-760-0087
Fax: 774-760-0088

Chicago
Itasca, IL
Tel: 630-285-0071
Fax: 630-285-0075

Dallas
Addison, TX
Tel: 972-818-7423
Fax: 972-818-2924

Detroit
Farmington Hills, MI
Tel: 248-538-2250
Fax: 248-538-2260

Kokomo
Kokomo, IN
Tel: 765-864-8360
Fax: 765-864-8387

Los Angeles
Mission Viejo, CA
Tel: 949-462-9523
Fax: 949-462-9608

Santa Clara
Santa Clara, CA
Tel: 408-961-6444
Fax: 408-961-6445

Toronto
Mississauga, Ontario,
Canada
Tel: 905-673-0699
Fax: 905-673-6509

ASIA/PACIFIC

Asia Pacific Office
Suites 3707-14, 37th Floor
Tower 6, The Gateway
Harbour City, Kowloon
Hong Kong
Tel: 852-2401-1200
Fax: 852-2401-3431

Australia - Sydney
Tel: 61-2-9868-6733
Fax: 61-2-9868-6755

China - Beijing
Tel: 86-10-8528-2100
Fax: 86-10-8528-2104

China - Chengdu
Tel: 86-28-8665-5511
Fax: 86-28-8665-7889

China - Fuzhou
Tel: 86-591-8750-3506
Fax: 86-591-8750-3521

China - Hong Kong SAR
Tel: 852-2401-1200
Fax: 852-2401-3431

China - Nanjing
Tel: 86-25-8473-2460
Fax: 86-25-8473-2470

China - Qingdao
Tel: 86-532-8502-7355
Fax: 86-532-8502-7205

China - Shanghai
Tel: 86-21-5407-5533
Fax: 86-21-5407-5066

China - Shenyang
Tel: 86-24-2334-2829
Fax: 86-24-2334-2393

China - Shenzhen
Tel: 86-755-8203-2660
Fax: 86-755-8203-1760

China - Shunde
Tel: 86-757-2839-5507
Fax: 86-757-2839-5571

China - Wuhan
Tel: 86-27-5980-5300
Fax: 86-27-5980-5118

China - Xian
Tel: 86-29-8833-7252
Fax: 86-29-8833-7256

ASIA/PACIFIC

India - Bangalore
Tel: 91-80-4182-8400
Fax: 91-80-4182-8422

India - New Delhi
Tel: 91-11-4160-8631
Fax: 91-11-4160-8632

India - Pune
Tel: 91-20-2566-1512
Fax: 91-20-2566-1513

Japan - Yokohama
Tel: 81-45-471- 6166
Fax: 81-45-471-6122

Korea - Daegu
Tel: 82-53-744-4301
Fax: 82-53-744-4302

Korea - Seoul
Tel: 82-2-554-7200
Fax: 82-2-558-5932 or
82-2-558-5934

Malaysia - Kuala Lumpur
Tel: 60-3-6201-9857
Fax: 60-3-6201-9859

Malaysia - Penang
Tel: 60-4-227-8870
Fax: 60-4-227-4068

Philippines - Manila
Tel: 63-2-634-9065
Fax: 63-2-634-9069

Singapore
Tel: 65-6334-8870
Fax: 65-6334-8850

Taiwan - Hsin Chu
Tel: 886-3-572-9526
Fax: 886-3-572-6459

Taiwan - Kaohsiung
Tel: 886-7-536-4818
Fax: 886-7-536-4803

Taiwan - Taipei
Tel: 886-2-2500-6610
Fax: 886-2-2508-0102

Thailand - Bangkok
Tel: 66-2-694-1351
Fax: 66-2-694-1350

EUROPE

Austria - Wels
Tel: 43-7242-2244-39
Fax: 43-7242-2244-393

Denmark - Copenhagen
Tel: 45-4450-2828
Fax: 45-4485-2829

France - Paris
Tel: 33-1-69-53-63-20
Fax: 33-1-69-30-90-79

Germany - Munich
Tel: 49-89-627-144-0
Fax: 49-89-627-144-44

Italy - Milan
Tel: 39-0331-742611
Fax: 39-0331-466781

Netherlands - Drunen
Tel: 31-416-690399
Fax: 31-416-690340

Spain - Madrid
Tel: 34-91-708-08-90
Fax: 34-91-708-08-91

UK - Wokingham
Tel: 44-118-921-5869
Fax: 44-118-921-5820

10/05/07

© 2007 Microchip Technology Inc.